

Assignment No.1

Based on General Concepts

1. Write a program to print "Welcome to C programming" on the screen.
2. Input a number and display its square and cube.
3. Input marks in three subjects and print total marks and percentage.
4. Input MRP from user and find the final price of product, when discount is 10%.
5. Accept temperature in fahrenheit and convert it into centigrade. [Formula: $C = (5 * (F - 32)) / 9$]
6. Accept the required values from user and calculate the area of rectangle, square and circle. Declare pi as constant.
7. Input amount, rate and principal amount from user and display simple interest and compound interest.
8. Accept rajat's basic salary through keyboard. His dearness allowance is 40% of basic allowance, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.
9. Write a program to exchange the values of two variables using and without using a third variable.

Based on Decision Construct

10. Write a program to find greater among two and three numbers.
11. Write a program to check whether a person is eligible to vote or not.
12. Write a program to check whether a number is even or odd.
13. Write a program to print grade of the student according to the aggregate marks entered by user. Assign grade as follows:

If marks are between 80-100, grade is A. 60-79, B grade. 40-59 C grade. Below 40, fail.

Based on Switch Case

14. Write a program to print name of weekday according to the number entered by user. E.g. If user enter 1, print Monday and so on.
15. Design a menu driven program to add, subtract, multiply and divide two numbers and print the result.

Based on Loops

16. Write a program to print counting from start to end. Accept values of start and end from user.
17. Write a program to print all even/odd numbers from start to end. Accept values of start and end from user.
18. Write a program to print table of a number. Accept number from user.
19. Input a number from user and find the sum of its digits.
20. Write a program to reverse a number.
21. Write a program to check whether a number is prime or not.
22. Write a program to check whether a number is palindrome or not.

Assignment No.2

Based on Arrays

1. Write a program to input array of ten numbers and print its sum.
2. Write a program to input marks of students of a class and find average marks. Input the value of number of students from user.
3. Write a program to find maximum and minimum number from a given set of values. Accept number of values from user.
4. Write a program to sort a list of numbers in ascending order. Ask the number of items from user.
5. Write a program to merge two arrays.
6. Design a menu driven program to add, subtract, multiply and divide two matrices. Ask dimensions (i.e. rows and columns) from user.
7. Write a program to find row sums and column sums of a matrix.
8. Write a program to find transpose of a matrix.
9. Write a program to search an element in the array using:
 - i) Linear search
 - ii) binary search.

Based on functions

10. Write a program to find square and cubes of 1 to 10 using functions.
11. Write a program to find sum of numbers between 1 and n and return the answer. Accept the value of n user and pass it to a function.
12. Write a program to check whether a number is even or odd by passing and without passing arguments. And return the answer.
13. Write a program to print sum of even numbers and sum of odd numbers from a sequence of numbers between start and end. Pass the values of start and end to function.
14. Write a program to compute factorial of a number with and without using recursion.
15. Write a program to print Fibonacci series with and without using recursion.
16. Write a program to interchange values of two variables by passing the arguments by value and by reference.
17. Write a program to demonstrate the use of command line arguments.

Assignment no. 3

Based on Pointers

1. Write a program to illustrate the concept of chain of pointers.
2. Write a program to illustrate the use of void pointers.
3. Write a program to illustrate the concept of pointer to an array.
 - a. Accessing complete array through pointers
 - b. Accessing a single element through pointers
 - c. Passing array to function
 - d. Accessing elements of 2-D array
4. Write a program to illustrate the concept of array of pointers.
5. Write a program to input ten values and find their sum using pointers.
6. Write a program to illustrate the concept of pointers to functions.
7. Write a program to input marks of some students. Accept the number of students from user. Print the list of original marks. If marks are less than 50, give a grace of 5 marks and print the modified marks.
8. Write a program to illustrate the concept of dynamic arrays (using malloc() and calloc()).
9. Write a program to illustrating the use of realloc().

Based on Strings

1. Design a menu driven program to implement following library functions:
 - a. strlen()
 - b. strcat()
 - c. strcpy()
 - d. strcmp()
 - e. strrev()
2. Design a menu driven program to implement following user defined functions (using functions):
 - a. strlen()
 - b. strcat()
 - c. strcpy()
 - d. strcmp()
 - e. strrev()
3. Write a program to check whether a string is palindrome or not (using user defined function).
4. Write a program to print all ASCII values.
5. Write a program to sort the list of names of students of a class.

Based on Series

Write programs for generating the sum of following series:

1. $x/1! + x/2! + x/3! + x/4! + \dots$
2. $1^1/x + 2^2/x + 3^3/x + 4^4/x + \dots$
3. $1^1/1! + 2^2/2! + 3^3/3! + 4^4/4! + \dots$
4. $1 + (1+2) + (1+2+3) + (1+2+3+4) + \dots$
5. $x - x^3/3! + x^5/5! + \dots$

Based on Preprocessor

1. Write a program to implement #define directive with and without argument.

Assignment no 4

Based on Structures and Union

1. Design a structure to store the details of 2 books.
2. Design a structure to input and print the details of 10 students of the class with fields: rollno, name, course, marks in 12th class.
3. Implement the concept of union to input and print the details of a product including id, name, price.

Based on Pyramids

1
23
456
78910..

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

A
BB
CCC
DDDD
EEEE

ABCDE
FGHI
JKL
MN
O

Based on Files

1. Write a program to read and write a string from and to a file.
2. Write a program to copy the contents of one file into another.
3. Write a program implementing the concept of unformatted I/O functions.